

$$= q_0 + q_1 \mathbf{i} + q_2 \mathbf{j} + q_3 \mathbf{k}$$

$$S^3$$

$$\omega = 2 \arccos(q_0)$$

$$|\mathbf{q}| = 1$$

$$(\phi_1, \Phi, \phi_2)$$

$$\mathbf{p} = \hat{n} \tan(\omega/2)$$

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$$SO(3)$$

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$$R(\phi_1, \Phi, \phi_2)$$

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